Asset Performance:

Ene 05 Cooling

Contributes to 40 credits available for asset energy rating

No Minimum Standard

Aim

To minimise operational energy consumption and the associated carbon emissions by increasing the intrinsic energy efficiency of the building fabric and the installed buildings services

Question

1. Is space cooling generated on-site?
2. What is the main system type for on-site cooling?
3. Please enter Energy Efficiency Ratio (EER) of the main on-site cooling generator, if known
4. Is the on-site cooling system central or local?
5. Is cooling distributed around the building via air?
6. In what year was the main chiller/cooling system replaced (if known)?
7. For district cooling only:
    a. Do you know what types of cooling generation are used for the network cooling supply?
    b. What are the main cooling generation sources?
    c. What is the percentage distribution losses for the cooling network (if known)?

Credits	Question	For questions 1, 2, 4, 5, 7a-b select a single answer option. For questions 3, 6, 7c enter value
-	1.	Yes / No
-	2.	a. Question not answered b. Asset is not cooled c. Localised (room) air conditioning unit d. Chiller e. On-site cogeneration f. Renewable cooling source g. Other (user defined)
-	3.	Enter EER of the cooling generator

Credits	Question	For questions 1, 2, 4, 5, 7a-b select a single answer option. For questions 3, 6, 7c enter value
-	4.	Central / Local
-	5.	Yes / No
-	6.	Enter year
-	7. a	Yes / No
-	7.b	Cooling source 1 Cooling source 2 Cooling source 3 Cooling source n
-	7.c	%

Methodology

Energy Efficiency Ratio (EER): CCHP/absorption chillers

To assess CCHP/absorption chillers using BREEAM In-Use, select the 'main system type for cooling' as 'Chiller'. The energy efficiency ratio (EER) for the cooling generator should then be calculated as follows:

EER = rated absorption chiller COP x 2

As an example, if the absorption chiller COP was 0.7, then the calculated energy efficiency ratio to be entered into the tool would be:

0.7 x 2 = 1.4

Where the calculated EER is below the minimum value that can be entered into the tool, please enter the minimum value. This correction is made to cancel out the primary energy factor normally applied for electrical chillers that would not be applicable in this instance. The COP value can be interchanged with the EER value.

Main cooling system type

Where there is more than one cooling system type in the asset, the Assessor must verify that the selected generation type is the main cooling source for the asset. This should be based upon respective system cooling output capacities.

Multiple cooling generation units for the main cooling system

If there is more than one cooling generation unit, the Assessor should calculate the average energy efficiency of the units based on their respective cooling generation capacities.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.

Criteria	Evidence requirement
1, 3, 4, 5	Visual inspection and verification through photographic evidence of the installed system(s).
All	Extract of relevant O&M manuals or a copy of manufacturer information.
6	Copy of documentation outlining when the cooling system was installed or replaced, such as, a) Extract of O&M manuals or a copy of manufacturer information for cooling generator/cooling system b) Service records c) Installation records d) Maintenance records
7	Copy of documentation provided by the district cooling operator.

Additional information

Asset energy calculator

In order to obtain the maximum possible number of credits for the asset energy performance as much information as possible should be entered. Where no value is entered, the asset energy rating calculator will assume a pessimistic default value.